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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA -2
Academic year	I – III Semester(2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520080035
Branch	AI&DS
Course Outcome	6

Case Study Topic : CloudSync Document Diff — detecting common edit history
A cloud-document collaboration platform compares two edited versions of a document to generate a compact change summary.
The backend uses the **Longest Common Subsequence (LCS)** algorithm to identify text portions that remained unchanged between versions.

Word Done Summary:

The **CloudSync Document Diff** project was developed to compare two edited versions of a document and generate a concise summary of the changes made. The system helps users quickly identify modifications without manually reviewing entire documents, making collaboration more efficient.

The project uses a document comparison algorithm to detect common and modified sections between two versions. By analyzing the edit history, it identifies added, deleted, and unchanged content, allowing users to understand document updates clearly and accurately.

To improve performance, the system focuses on finding the longest common sequences between document versions and highlighting only the differences. This reduces processing time and generates a compact change summary even for large documents with multiple revisions.

Overall, the project successfully provides an efficient document comparison solution for cloud-based collaboration platforms. It enhances version tracking, simplifies change management, and improves productivity by presenting document edits in an organized and easy-to-understand format.

Implementation Details:

1. **Document Input Processing** – Loaded and preprocessed two document versions for comparison.

2. **Difference Detection** – Compared the documents to identify added, deleted, and modified content.

3. **Common Sequence Analysis** – Used a matching algorithm to find common text between versions.

4. **Change Summary Generation** – Generated a compact and clear summary of all detected edits.

OutPut:

```
b.deepak@BDeepaks-MacBook-Air DSA 2 % javac LCSDemo.java
b.deepak@BDeepaks-MacBook-Air DSA 2 % java LCSDemo
String A: NETWORK
String B: NOTEBOOK
LCS DP Table:
0 0 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1 1
0 1 1 1 2 2 2 2 2
0 1 1 2 2 2 2 2 2
0 1 1 2 2 2 2 2 2
0 1 2 2 2 2 3 3 3
0 1 2 2 2 2 3 3 3
0 1 2 2 2 2 3 3 4

Longest Common Subsequence: NEOK
LCS Length: 4
b.deepak@BDeepaks-MacBook-Air DSA 2 %
```

GitHub Link:https://github.com/bdeepakg-boop/DSA-Project/upload/main

Student Signature	Faculty Signature
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